Case Study: CAR0001

Cardiology — Chest Pain

SOAP Extraction Analysis · GPT-5.6 Terra

Clinical Fidelity	Logical Consistency	SOAP Structural Fidelity	Overall Average
95%	95%	90%	93.3%

Evaluated by gemini-3.5-flash-lite | Scale: 0–100% | Paper-aligned metrics

CAR0001 — Speech-to-Text Transcript Comparison

Ground Truth dialogue vs GPT-5.6 Terra diarized STT output (first ~100 words)

📄 Real Transcript (Ground Truth)

D: What brought you in today?
P: Sure, I'm I'm just having a lot of chest pain and and so I thought I should get it checked out.
D: OK, before we start, could you remind me of your gender and age?
P: Sure 39, I'm a male.
D: OK, and so when did this chest pain start?
P: It started last night, but it's becoming sharper.
D: OK, and where is this pain located?
P: It's located on the left side of my chest.
...

🤖 GPT-5.6 Terra Diarized STT

P: brought you in today? Sure I'm just having a lot of chest pain and so I thought I should get it
D: checked out. Okay and before we start could you remind me of your gender and age? Sure I'm 39
P: I'm a male. Okay and so when did this chest pain start? It started last night but it's becoming
D: sharper. Okay and where is this pain located? It's located on the left side of my chest. Okay and
...

CAR0001 — Clinical Summary Comparison

AI-extracted clinical summary: Ground Truth vs GPT-5.6 Terra (first ~100 words)

📄 Ground Truth Summary

The patient, a 39-year-old male, presents with 8 hours of sharp, constant left-sided chest pain that worsens with lying down and deep breaths. He reports associated symptoms of lightheadedness, dyspnea, and mild palpitations, and noted visible neck swelling without accompanying pain.

🤖 GPT-5.6 Terra Summary

A 39-year-old man presents with acute, constant, sharp left-sided chest pain beginning last night (approximately 8 hours), rated 7-8/10 and becoming sharper. Pain is worse when lying down and possibly with deep inspiration, and is associated with dyspnea, light-headedness, mild palpitations, sweating with dyspnea, intermittent noisy breathing during severe dyspnea, and perceived neck swelling. He has significant cardiovascular risk factors including cigarette smoking and a father with myocardial infarction at age 45.

CAR0001 — Structured SOAP Report Comparison

SOAP Field	Ground Truth	Predicted (GPT-5.6 Terra)
Chief Complaint	Chest pain	Acute left-sided chest pain.
Symptoms	sharp left-sided chest pain | worsening pain on lying down | worsening pain on deep breath | lightheadedness | dyspnea | palpitati...	Constant sharp left-sided chest pain, rated 7-8/10 and becoming sharper | Pain worsened by lying down | Pain possibly worsened by ...
Symptom Onset	Last night (approximately 8 hours ago)	Started last night; patient estimates duration of approximately 8 hours. Symptoms began while moving furniture, without perceived ...
Medical History	—	No prior medical conditions | No recent hospitalizations | No prior surgeries | No previous similar episode | Immunizations report...
Curr. Medications	—	—
Allergies	none	No known medication allergies
Exam Findings	reported visible neck swelling | patient reports 7-8/10 pain severity	Patient reports that neck appears mildly swollen
Diagnosis	—	Acute left-sided chest pain, unspecified etiology
Diff. Diagnosis	Myocardial infarction (due to family history and symptoms) | Pericarditis (suggested by positional component) | Pneumothorax | Mus...	—
Prescribed Meds	—	—
Treatment Plan	—	—
Investigations	—	—
Follow Up	—	—

Case Study: MSK0005

Musculoskeletal — Elbow Pain

SOAP Extraction Analysis · GPT-5.6 Terra

Clinical Fidelity	Logical Consistency	SOAP Structural Fidelity	Overall Average
98%	100%	95%	97.7%

Evaluated by gemini-3.5-flash-lite | Scale: 0–100% | Paper-aligned metrics

MSK0005 — Speech-to-Text Transcript Comparison

Ground Truth dialogue vs GPT-5.6 Terra diarized STT output (first ~100 words)

📄 Real Transcript (Ground Truth)

D: What brings you in?
P: Hi, I've I've had this pain on the outside of my uh right elbow now it's it I first started knowing, noticing it several months ago, but recently it's just been more painful.
D: OK, so you said several months ago. Um did anything happen several months ago? Was there any sort of trigger, trauma, anything like that to that area?
...

🤖 GPT-5.6 Terra Diarized STT

P: Hi, I've had this pain on the outside of my right elbow now. I first started noticing
P: it several months ago, but recently it's just been more painful.
D: Okay, so you said several months ago. Did anything happen several months ago? Was there
D: any sort of trigger, trauma, anything like that to that area?
P: No, there wasn't any trauma or any triggers that I noticed. I was just feeling it a bit
P: more at the end of work. Yeah, I was just feeling the pain a bit more at the end of
P: work.
...

MSK0005 — Clinical Summary Comparison

AI-extracted clinical summary: Ground Truth vs GPT-5.6 Terra (first ~100 words)

📄 Ground Truth Summary

The patient is a healthy adult presenting with persistent right lateral elbow pain for several months, worsening with repetitive wrist movements at work. Examination findings are consistent with lateral epicondylitis, evidenced by pain on palpation of the lateral epicondyle and pain with resisted wrist extension and flexion. The patient was advised on activity modification, use of compression sleeves, continued over-the-counter analgesics, and physical therapy for recovery.

🤖 GPT-5.6 Terra Summary

Patient reports progressively worsening, localized right lateral elbow pain for several months, associated with repetitive line work and aggravated by wrist movement and lifting. Findings are consistent with lateral epicondylitis (tennis elbow), with preserved range of motion and no trauma, neurologic symptoms, inflammatory signs, or systemic symptoms.

MSK0005 — Structured SOAP Report Comparison

SOAP Field	Ground Truth	Predicted (GPT-5.6 Terra)
Chief Complaint	Right elbow pain	Pain on the outside of the right elbow.
Symptoms	pain on the outside of the right elbow | pain worse with wrist bending | pain with lifting heavy objects | aching pain | sharp pai…	Localized right lateral elbow pain for several months, recently worsening | Pain is worse at the end of work | Achy pain with inte…
Symptom Onset	Several months ago	Several months ago; no identified trauma or trigger. Pain has progressively worsened over the past several months.
Medical History	C-section (17 years ago)	Otherwise healthy; no longstanding medical conditions reported | Cesarean section for son approximately 17 years ago; not an emerg…
Curr. Medications	Tylenol | Advil	Acetaminophen (Tylenol) as needed for pain | Ibuprofen (Advil) as needed for pain | Over-the-counter topical pain ointment, unspec…
Allergies	No known allergies	No known allergies
Exam Findings	No redness or swelling of the elbow | No deformity or misalignment | Pain on palpation of the lateral aspect of the right elbow | …	Patient reports full range of motion of the right elbow and hand | No visible right elbow redness, swelling, skin changes, deformi…
Diagnosis	Lateral epicondylitis (Tennis elbow)	Right lateral epicondylitis (tennis elbow), likely related to repetitive occupational strain and continuous wrist extensor muscle …
Diff. Diagnosis	—	Right elbow fracture, considered unlikely due to absence of trauma
Prescribed Meds	Continue Tylenol or NSAIDs as needed	—
Treatment Plan	Activity modifications (avoid repetitive strain/extensor muscle use at work) | Use of compression sleeve | Physical therapy | Appl…	Modify work activities and avoid overly repetitive tasks, particularly those using the forearm extensor muscles | Use a compressio…
Investigations	—	No imaging recommended; no trauma history and clinician has low concern for fracture
Follow Up	The condition should resolve in about six months to two years at most, but normally resolves in a couple of months with physical t…	—

Case Study: RES0050

Respiratory — Chronic Cough

SOAP Extraction Analysis · GPT-5.6 Terra

Clinical Fidelity	Logical Consistency	SOAP Structural Fidelity	Overall Average
98%	95%	95%	96.0%

Evaluated by gemini-3.5-flash-lite | Scale: 0–100% | Paper-aligned metrics

RES0050 — Speech-to-Text Transcript Comparison

Ground Truth dialogue vs GPT-5.6 Terra diarized STT output (first ~100 words)

📄 Real Transcript (Ground Truth)

D: I was wondering if you could tell me what brought you in.
P: Sure, um, I have been coughing a lot. I've been feeling short of breath and I just feel really wheezy, so I thought I should come check it out.
D: OK, when did the cough start?
P: I've always had a cough. Well, I guess I've had a cough for the past five years, but over the past three days it's gotten a lot worse.
D: OK, did anything happen 3 days ago that has brought the cough on?
...

🤖 GPT-5.6 Terra Diarized STT

D: I was wondering if you could tell me what brought you in?
P: Sure. I have been coughing a lot. I've been feeling short of breath, and I just feel really
P: wheezy, so I thought I should come check it out.
D: Okay. When did the cough start?
P: I've always had a cough. Well, I guess I've had a cough for the past five years,
P: but over the past three days, it's gotten a lot worse.
D: Okay. Did anything happen three days ago that has brought the cough on?
...

RES0050 — Clinical Summary Comparison

AI-extracted clinical summary: Ground Truth vs GPT-5.6 Terra (first ~100 words)

📄 Ground Truth Summary

A patient with a 5-year history of chronic cough presented for evaluation of worsening cough, increased yellow/green sputum, wheezing, and exertional shortness of breath over the past 3 days. The patient has a 45 pack-year smoking history (quit 2 years ago) and a past medical history of hypertension and diabetes. The differential diagnosis includes viral infection, bacterial infection, COPD exacerbation, and COVID-19, with plans initiated for a COVID-19 PCR nasal swab and chest X-ray.

🤖 GPT-5.6 Terra Summary

Patient presents with a chronic cough for five years that has worsened substantially over the past three days, with increased yellow/green sputum, wheezing since last night, exertional shortness of breath, fatigue, and dull chest discomfort only after coughing. The doctor considers viral or bacterial respiratory infection, COPD exacerbation, and COVID-19, and plans COVID PCR testing, chest radiography, and additional investigations.

RES0050 — Structured SOAP Report Comparison

SOAP Field	Ground Truth	Predicted (GPT-5.6 Terra)
Chief Complaint	Worsening cough, shortness of breath, and wheezing over the past 3 days.	Worsening cough with shortness of breath and wheezing.
Symptoms	Worsening cough (chronic for 5 years, exacerbated past 3 days, worse in the morning) | Shortness of breath (worse with walking and...	Chronic cough for approximately 5 years, markedly worse over the past 3 days | Cough worse in the morning | Increased sputum produ...
Symptom Onset	3 days ago for acute exacerbation of symptoms; chronic cough present for 5 years.	Baseline cough for approximately 5 years. Cough and sputum production worsened 3 days ago after returning home from grocery shoppi...
Medical History	Hypertension | Diabetes	Hypertension | Diabetes | No prior surgeries | No medication allergies | Denies ear pain, eye pain, ear/eye discharge, rhinorrhea,...
Curr. Medications	Beta-blocker | Metformin | Calcium supplement | Vitamin D	Beta blocker (unspecified) for hypertension | Metformin for diabetes | Calcium supplement | Vitamin D supplement
Allergies	—	No known medication allergies
Exam Findings	—	No physical examination findings documented in the transcript.
Diagnosis	—	Acute worsening of chronic productive cough with wheezing and exertional shortness of breath; etiology under investigation
Diff. Diagnosis	Viral respiratory infection | Bacterial respiratory infection | COPD exacerbation | COVID-19 infection	Viral respiratory infection | Bacterial respiratory infection | COPD exacerbation | COVID-19
Prescribed Meds	—	—
Treatment Plan	Diagnostic workup to determine etiology of acute respiratory exacerbation.	Arrange pneumococcal vaccination with family doctor, as recommended for adults older than 65 in Ontario to protect against certain...
Investigations	COVID-19 PCR nasal swab | Chest X-ray | Other laboratory/diagnostic investigations	COVID-19 PCR nasal swab | Chest X-ray | Other unspecified investigations
Follow Up	Set up pneumococcal vaccination with family doctor; follow up for investigation results to get to the bottom of shortness of breat...	No specific return precautions or follow-up timeframe were provided; patient was advised to arrange pneumococcal vaccination with ...

Case Study: GAS0003

Gastroenterology — Abdominal Pain

SOAP Extraction Analysis · GPT-5.6 Terra

Clinical Fidelity	Logical Consistency	SOAP Structural Fidelity	Overall Average
95%	95%	90%	93.3%

Evaluated by gemini-3.5-flash-lite | Scale: 0–100% | Paper-aligned metrics

GAS0003 — Speech-to-Text Transcript Comparison

Ground Truth dialogue vs GPT-5.6 Terra diarized STT output (first ~100 words)

📄 Real Transcript (Ground Truth)

D: What brings you in here today?
P: Yeah hi, I'm I I have a pain in my belly. It started a couple hours ago.
D: OK, and where exactly in your belly do you have this pain?
P: It's mostly up here on the right side.
D: The right side, OK and can you describe what kind of pain is it? Is it like a dull pain or sharp pain, an aching pain?
...

🤖 GPT-5.6 Terra Diarized STT

D: What brings you in here today?
P: Yeah.
P: Hi.
P: I have a pain in my belly.
P: It started a couple of hours ago.
D: Okay.
D: And where exactly in your belly do you have this pain?
P: It's mostly up here on the right side.
D: On the right side?
P: Yeah.
D: Okay.
D: And can you describe what kind of pain is it?
D: Is it like a dull pain, a sharp pain, an aching pain?
P: I guess it's an aching pain, I'd say.
...

GAS0003 — Clinical Summary Comparison

AI-extracted clinical summary: Ground Truth vs GPT-5.6 Terra (first ~100 words)

📄 Ground Truth Summary

Patient is presenting with a constant, aching right-sided abdominal pain that has persisted for 6 hours after an initial 2-hour period of intermittent pain. The patient reports a history of similar postprandial discomfort associated with fatty meals and is currently experiencing nausea but no vomiting, fevers, or bowel changes.

🤖 GPT-5.6 Terra Summary

Patient presents with severe, worsening right upper abdominal aching pain that began about eight hours ago, initially intermittent but constant for the past five to six hours. Pain is rated 8/10, worsens with eating, and is associated with mild nausea, reduced food intake, and mild fatigue; the patient reports similar intermittent postprandial episodes after fatty foods in the past.

GAS0003 — Structured SOAP Report Comparison

SOAP Field	Ground Truth	Predicted (GPT-5.6 Terra)
Chief Complaint	right-sided abdominal pain	Right upper abdominal pain.
Symptoms	aching pain in right side of abdomen | nausea | fatigue | decreased appetite	Right upper abdominal aching pain, non-radiating, rated approximately 8/10 | Pain began about 8 hours ago while preparing to sleep...
Symptom Onset	8 hours ago	About 8 hours before the consultation; initially intermittent for approximately 2 hours, then constant for the past 5-6 hours.
Medical History	diabetes	Diabetes | No prior hospitalizations | Wisdom teeth extraction as a teenager | Family history: father had a heart attack in his 60...
Curr. Medications	metformin	Metformin | Acetaminophen (Tylenol), tried for current pain without relief
Allergies	—	No known medication allergies | No known environmental allergies
Exam Findings	—	—
Diagnosis	—	Right upper quadrant abdominal pain
Diff. Diagnosis	indigestion	—
Prescribed Meds	—	—
Treatment Plan	—	—
Investigations	—	—
Follow Up	—	—

Case Study: DER0001

Dermatology — Skin Rash

SOAP Extraction Analysis · GPT-5.6 Terra

Clinical Fidelity	Logical Consistency	SOAP Structural Fidelity	Overall Average
98%	95%	92%	95.0%

Evaluated by gemini-3.5-flash-lite | Scale: 0–100% | Paper-aligned metrics

DER0001 — Speech-to-Text Transcript Comparison

Ground Truth dialogue vs GPT-5.6 Terra diarized STT output (first ~100 words)

📄 Real Transcript (Ground Truth)

P: Hey Doc, I'm a 45 year old man coming in, I'm worried there's a little old rash on my leg that I'm getting nervous about.
D: Hi, so you I understand you have a rash on your leg.
P: I don't really know what I would call it. That's what I told the nurse when I came in, but it more hurts than anything.
D: OK, uh, how long is this been going on for?
...

🤖 GPT-5.6 Terra Diarized STT

P: Hey doc, I'm a 45 year old man coming in. I'm worried there's a little rash on my leg I'm getting nervous about.
D: Hi. So I understand you have a rash on your leg.
P: I don't really know what I would call it. That's what I told the nurse when I came in, but it's more hurts than anything.
D: Okay. How long has this been going on for?
P: It's been, it's been looking weird for about a week, but only, yes, it started hurting yesterday.
...

DER0001 — Clinical Summary Comparison

AI-extracted clinical summary: Ground Truth vs GPT-5.6 Terra (first ~100 words)

📄 Ground Truth Summary

A 45-year-old male with poorly controlled diabetes presents with a painful, swollen, and red rash on his right ankle of one week's duration, now with purulent drainage. Patient also reports constitutional symptoms including fatigue, increased hunger, and a subjective feeling of fever over the last 12 hours.

🤖 GPT-5.6 Terra Summary

45-year-old man with poorly controlled diabetes presents with a one-week abnormal right ankle lesion that became painful yesterday, with increasing swelling, spreading redness, warmth, an open wound from a scab, and white drainage. He reports feeling hot for 12 hours, recent malaise and low energy, polyuria, increased hunger, and chronic reduced sensation in both feet; examination and discussion with the supervising doctor were planned.

DER0001 — Structured SOAP Report Comparison

SOAP Field	Ground Truth	Predicted (GPT-5.6 Terra)
Chief Complaint	Painful, strange-looking rash on right ankle.	Painful, swollen, red, abnormal-looking lesion/rash on the right ankle.
Symptoms	rash on right ankle | pain | swelling | erythema | scabbing | purulent drainage | subjective fever/feeling hot | fatigue | increas…	Right ankle lesion looked abnormal for about one week | Right ankle pain started yesterday | Increasing swelling and enlargement o…
Symptom Onset	Rash began 1 week ago, worsening/painful since yesterday, purulent drainage started within last 12 hours.	Right ankle lesion appeared abnormal about one week ago; pain began yesterday. Swelling and white drainage began with the swelling…
Medical History	Diabetes mellitus (poorly controlled, A1C ~9) | Hypercholesterolemia | History of asthma | Chronic back pain | History of sinusiti…	Type 2 diabetes mellitus, poorly controlled; stopped checking blood glucose and reports last HbA1c was approximately 9% | Hypercho…
Curr. Medications	Rosuvastatin | Metformin | Two other unspecified diabetes medications	Rosuvastatin | Metformin | Two additional diabetes medications, names unknown; patient plans to provide medication list | Acetamin…
Allergies	Penicillin (reaction: facial bloating, respiratory distress)	Penicillin: childhood reaction with facial swelling and difficulty breathing
Exam Findings	Right ankle appears swollen and erythematous | Open wound/leaking of purulent material present | Right leg feels warmer to the tou…	Patient reports visible swelling of the right ankle lesion, with enlargement compared with prior appearance | Patient reports redn…
Diagnosis	—	Suspected right ankle cellulitis/infected skin wound in the setting of poorly controlled diabetes | Poorly controlled type 2 diabe…
Diff. Diagnosis	Infected diabetic ulcer | Cellulitis	—
Prescribed Meds	—	—
Treatment Plan	Physical examination of the leg | Consultation with supervising physician	Perform closer physical examination of the right leg/ankle lesion | Clinician to discuss the case with the supervising doctor afte…
Investigations	—	—
Follow Up	Patient to undergo physical examination of the leg immediately.	Immediate next step was physical examination of the leg followed by discussion with the supervising doctor to determine further ma…